Rare-miss growth: a finite-cluster law and certified bounds

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Abstract

We study products of fixed strictly positive matrices when every window of k factors contains at most m occurrences of a designated miss mode. For fixed dimension and m, a uniform cluster-packing argument identifies the first-order logarithmic growth correction as the largest Perron-weighted gain of a cluster with at most m misses. The argument covers all admissible words. A realizable finite-gap cutoff then gives certified two-sided bounds at finite k, an effective accuracy threshold, and an $O(\log k/k)$ error bound for the scaled coefficient. We formalize the main law and the finite-window implications in Lean, with explicit correspondence between the mathematical and formal definitions. Examples explain why no dimension-independent list of pair gaps suffices and why the lower-bound periodic witnesses need not be finite-window optimizers.

Draft status. The main law and the conditional finite-window implications have kernel-checked Lean proofs. The automatic minorization-tail construction and the additional example propositions are not all formalized. Publication priority, human specialist review, and author metadata remain pending. The manuscript and proof development used AI assistance.

1 Introduction and scope

A sliding-window loss rule controls local bursts, not only average loss frequency. We ask how worst-case product growth changes when the window becomes large while the matrices and the allowed number of misses remain fixed. Periodic products supply lower bounds, but the central difficulty is a matching estimate uniform over every admissible word.

The key observation is combinatorial. Join nearby misses into clusters and assign each cluster to a bin determined by its first miss. With bin length $k - mq$, the original k-window rule bounds the *total* number of misses in all clusters assigned to one bin by m. Perron separation converts the matrix product into cluster gains; a joining argument bounds their product within each bin by one finite-cluster supremum. Charging each miss independently would lose this budget-sharing information.

The fixed-language setting has established spectral and multinorm representations [7, 6, 10]. In particular, Kozyakin–Kuznetsov–Chebotarev [6, Sections 3.3–3.5] recodes block-frequency constraints as a finite-type language and obtains periodic spectral representation. Those results leave the cycle supremum to be evaluated as the window varies. Weakly-hard control also has scalar rate abstractions [2] and nonlinear sufficient stability criteria [4]; these are not an equality for the present matrix growth coefficient. Conditional-average variational results [12] concern a different constraint.

Rank-one identities and maximum-cycle methods are established [1, 8]. Perron pairings also occur in slow-switching expansions [9], where retained environmental durations are stretched under

a specified periodic or random law. Here short miss clusters coexist with a hard adversarial window constraint. Related hybrid singular-perturbation results [3] do not directly supply that packing comparison.

We prove the cluster law, then obtain a quantitative strengthening by clamping every long gap to a fixed *realizable* gap. The resulting maximum is attained by a finite word, providing an effective lower witness as well as an upper enclosure. The Lean development verifies the main law, the cutoff comparison, both finite-window inequalities, and an all-later-windows accuracy implication. Section 7 separates imported results, checked statements and classical-only steps.

The scope is fixed strictly positive discrete matrices and fixed miss budget. We make no claim for arbitrary signed modes, growing budgets, nonlinear plants or continuous intersample safety, and do not assume that every matrix pair is realizable by a particular feedback architecture. The examples explain limitations of the general conclusion. Publication novelty remains provisional; the unclosed source and human-review gates are listed in Section 8.

2 The cluster law

Fix integers $d, m \geq 1$ and entrywise strictly positive matrices $C, D \in \mathbb{R}^{d\times d}$. For integers $k > m$, let

$$\Sigma_{m,k} = \left\{ \sigma \in \{0,1\}^{\mathbb{N}_0} : \sum_{i=t}^{t+k-1} (1-\sigma_i) \leq m \text{ for every } t \in \mathbb{N}_0 \right\}.$$

The symbol 1 denotes success and 0 denotes a miss. Write $A_1 = C$, $A_0 = D$, and define the constrained growth rate

$$\rho_{m,k} = \limsup_{N\to\infty} \sup_{\sigma\in\Sigma_{m,k}} \left\| A_{\sigma_{N-1}} \cdots A_{\sigma_0} \right\|^{1/N}. \tag{1}$$

For definiteness, take the elementwise maximum norm, as in the formal statement. Standard finite-dimensional norm equivalence shows that any other fixed matrix norm gives the same rate. Put

$$\lambda = \operatorname{spr}(C), \qquad B = C/\lambda, \qquad E = D/\lambda.$$

Perron–Frobenius theory gives positive vectors r, ℓ such that

$$Br = r, \qquad \ell^{\mathsf{T}} B = \ell^{\mathsf{T}}, \qquad \ell^{\mathsf{T}} r = 1, \qquad B^n \longrightarrow P := r\ell^{\mathsf{T}}. \tag{2}$$

For $j \geq 2$ and gaps $\boldsymbol{g} \in \mathbb{N}_0^{j-1}$ define

$$K_j(\boldsymbol{g}) = EB^{g_{j-1}} E \cdots B^{g_1} E, \qquad K_1 = E.$$

Thus g_1 is the number of successes after the first miss in chronological order; matrices multiply in the opposite order. Define

$$S_j = \sup_{\boldsymbol{g}\in\mathbb{N}_0^{j-1}} \ell^{\mathsf{T}} K_j(\boldsymbol{g}) r, \qquad M_m = \max\{1, S_1, \ldots, S_m\}. \tag{3}$$

For $j = 1$ the supremum has no gap variables. Each S_j is positive and finite because the powers of B are bounded and j is fixed. The supremum need not be attained at finite gaps.

Theorem 1 (Fixed-budget rare-miss law)**.** *With C, D, d, m fixed as above,*

$$\lim_{k\to\infty} k \log \frac{\rho_{m,k}}{\lambda} = \log M_m. \tag{4}$$

Equivalently, $\log \rho_{m,k} = \log\lambda + k^{-1} \log M_m + o(k^{-1})$.

The all-success sequence explains the maximum with 1 in (3). In particular, the correction is nonnegative. The theorem does not identify an exactly maximizing word at each finite window size.

3 Proof of Theorem 1

3.1 Uniform cluster packing

Choose an integer $q = q(k)$ with $q \to \infty$ and $q/k \to 0$, for instance $q = \lfloor \sqrt{k} \rfloor$. Take k large enough that $mq + 1 \le k$, and set $L = k - mq > 0$. In a finite prefix of an admissible sequence, join successive misses into the same cluster whenever fewer than q successes lie between them.

Lemma 2 (Window packing)**.** *Each cluster has at most m misses and spans at most mq slots. Assign a cluster to the integer bin $[tL, (t+1)L - 1]$ containing its first miss. All clusters assigned to one bin contain at most m misses in total.*

Proof. If one cluster contained $m + 1$ misses, the span from the first to the last, including both endpoints, would be at most

$$m(q-1) + (m+1) = mq + 1 \le k,$$

contradicting the budget. A cluster of at most m misses spans no more than $(m-1)q + 1 \le mq$ slots.

A cluster starting in bin t ends no later than

$$(t+1)L - 1 + (mq - 1) = tL + k - 2 < tL + k.$$

Consequently every miss in every cluster assigned to bin t belongs to the same k-window $[tL, tL + k - 1]$. That window has at most m misses. If it extends beyond the chosen prefix, use the original infinite admissible sequence. Truncating the final cluster of a prefix does not increase its span or count. □

3.2 Perron separation and joining

Use the vector norm $\|x\|_r = \max_i |x_i|/r_i$ and its induced matrix norm. For every nonnegative matrix W,

$$\|W\|_r = \max_i \frac{(Wr)_i}{r_i} \le \kappa \ell^{\mathsf{T}} W r, \qquad \kappa = \frac{1}{\min_i \ell_i r_i}. \tag{5}$$

Indeed, the scalar on the right before multiplication by κ is a weighted average of these ratios with positive weights $\ell_i r_i$ summing to 1. The induced-norm identity follows from nonnegativity, with equality attained at $x = r$.

For $q \ge 1$ put

$$\varepsilon_q = \sup_{g \ge q} \max_{a,b} \frac{|(B^g - P)_{ab}|}{P_{ab}}. \tag{6}$$

All entries of P are positive. By (2), ε_q is finite, tends to zero, and gives the entrywise bound

$$B^g \le (1 + \varepsilon_q) P \qquad (g \ge q). \tag{7}$$

Lemma 3 (Joining cluster gains)**.** *Let $K^{(1)}, \dots, K^{(s)}$ be finite clusters containing a total of j misses. Then*

$$\prod_{i=1}^{s} \ell^{\mathsf{T}} K^{(i)} r \le S_j.$$

In particular, the product of gains of any collection with at most m total misses is at most M_m.

Proof. Insert h successes between consecutive clusters. For each finite h the result is a j-miss cluster, so its gain is at most S_j. Let $h \to \infty$. Every inserted B^h converges to P, and $PWP = (\ell^{\mathsf{T}} W r)P$ factors the limiting gain. This argument uses a supremum, not attainment of a maximizing gap tuple. □

3.3 The uniform matrix-product bound

Let W be a normalized length-N product and let z be its number of misses. Disjoint full k-blocks and one final fragment give

$$z \le m\lfloor N/k \rfloor + m. \tag{8}$$

The final fragment also has at most m misses because it is contained in a k-window of the infinite extension.

Different clusters are separated by at least q successes. Replace each such success run using (7). Nonnegative matrix multiplication preserves entrywise inequalities. In the scalar $\ell^{\mathsf{T}} W r$, the outer success runs vanish by (2), and the inserted rank-one separators factor the remaining expression into cluster gains. There are at most z separators. By Lemmas 3 and 2, each occupied bin contributes at most M_m. At most $\lceil N/L \rceil$ bins are occupied. Thus, including the zero-miss case,

$$\|W\|_r \le \kappa (1 + \varepsilon_q)^z M_m^{\lceil N/L \rceil}. \tag{9}$$

This is a bound on every admissible word.

Fix k, q first. Taking suprema, Nth roots and the long-product limit superior in (9) yields

$$\log \frac{\rho_{m,k}}{\lambda} \le \frac{\log M_m}{k - mq} + \frac{m}{k} \log(1 + \varepsilon_q). \tag{10}$$

Multiply by k and then let $k \to \infty$. Since $q/k \to 0$ and $\varepsilon_q \to 0$,

$$\limsup_{k \to \infty} k \log \frac{\rho_{m,k}}{\lambda} \le \log M_m. \tag{11}$$

3.4 Matching periodic lower witnesses

Fix a finite cluster W with $j \le m$ misses, length a, and gain $s = \ell^{\mathsf{T}} W r > 0$. Repeat this cluster followed by k successes. The period is $k + a \ge k$. Residues modulo the period are distinct within any k-window, so that window has at most j misses. The sequence is therefore admissible. Its period matrix is $Q_k = B^k W$, and

$$Q_k r \longrightarrow PWr = sr.$$

For every $0 < t < s$, positivity of r gives $Q_k r \ge tr$ for all sufficiently large k. Nonnegative multiplication then gives $Q_k^N r \ge t^N r$. Using the norm/gain comparison and the repeated-period subsequence in the rate definition yields

$$\rho_{m,k}/\lambda \ge t^{1/(k+a)}, \qquad \liminf_{k \to \infty} k \log(\rho_{m,k}/\lambda) \ge \log t.$$

Let t increase to s, take the supremum over each fixed finite cluster, and include the all-success lower bound. The result is $\log M_m$. Together with the uniform upper bound, this proves Theorem 1. No varying maximizing word is interchanged with the k-limit, and no spectral-radius continuity theorem is needed for this argument.

4 Certified finite-window bounds

The asymptotic law does not say how large k must be. A two-sided Perron tail makes the approximation effective. Fix integers $m, q \geq 1$ and a certified $0 \leq \varepsilon < 1$ such that, entrywise,

$$(1-\varepsilon)P \leq B^g \leq (1+\varepsilon)P \qquad (g \geq q). \tag{12}$$

Put $R = (1+\varepsilon)/(1-\varepsilon)$ and define the genuine finite maximum

$$A_m(q) = \max\left\{1,\ \max_{1\leq j\leq m}\ \max_{\boldsymbol{g}\in\{0,\dots,q\}^{j-1}} \ell^{\mathsf{T}} K_j(\boldsymbol{g}) r\right\}. \tag{13}$$

Unlike a search containing the ideal projector P, every candidate here is realizable by a finite word. There are $1 + \sum_{j=1}^{m}(q+1)^{j-1}$ candidates before duplicate values are removed.

Lemma 4 (Attained cutoff)**.** *Under* (12)*,*

$$1 \leq A_m(q) \leq M_m \leq R^{m-1} A_m(q).$$

A maximizing word can be chosen with at most m misses and length at most $m+(m-1)q$, allowing the empty word for the value 1*.*

Proof. For $g \geq q$, the two tail inequalities imply $B^g \leq (1+\varepsilon)P \leq RB^q$. For $g \leq q$, nonnegativity and $R \geq 1$ give $B^g \leq RB^g$. Clamp each gap to $\min(g, q)$ and multiply these entrywise inequalities through nonnegative factors. A j-miss cluster has $j-1$ gaps, hence gain at most $R^{j-1}A_m(q)$. The empty-word contribution is also bounded by $R^{m-1}A_m(q)$. Taking suprema gives the upper comparison. Every cutoff candidate is already allowed in M_m, giving the lower one. The finite maximum is attained, and a j-miss candidate has length at most $j + (j-1)q$. □

Write $\Xi_{m,k} = k \log(\rho_{m,k}/\lambda)$.

Theorem 5 (Two-sided finite-window enclosure)**.** *Suppose* (12) *holds and $k \geq m(q+1)$. Then*

$$\max\{0, \log A_m(q) + \log(1-\varepsilon)\} \leq \Xi_{m,k}, \tag{14}$$

$$\Xi_{m,k} \leq \frac{k}{k-mq}\big(\log A_m(q) + (m-1)\log R\big) + m\log(1+\varepsilon). \tag{15}$$

Proof. Choose an attaining word W of length $a \leq m+(m-1)q$. The hypothesis ensures $k-a \geq q$. Repeat W followed by $k-a$ successes. Each k-window is one period and contains at most m misses. Its period matrix $Q = B^{k-a}W$ satisfies

$$Qr \geq (1-\varepsilon)PWr = (1-\varepsilon)A_m(q)r.$$

Iteration, the norm/gain comparison and the rate definition yield (14), together with the all-success sequence. For the upper bound, (10) and Lemma 4 give (15). In particular, the upper bound applies to all admissible products, not only the lower witness. □

4.1 Error and a certified accuracy threshold

Let $H = \log M_m \geq 0$. The proof also gives

$$-\big((m-1)\log R - \log(1-\varepsilon)\big) \leq \Xi_{m,k} - H \leq \frac{mq}{k-mq} H + m\log(1+\varepsilon). \tag{16}$$

For $\varepsilon \leq 1/2$, $k \geq \max\{m(q+1), 2mq\}$, and any verified $\overline{H} \geq H$, the elementary logarithmic inequalities imply

$$-(3m-1)\varepsilon \leq \Xi_{m,k} - H \leq \frac{2mq\,\overline{H}}{k} + m\varepsilon. \tag{17}$$

Indeed $\log(1+\varepsilon) \leq \varepsilon$ and $-\log(1-\varepsilon) \leq 2\varepsilon$.

Given $\delta > 0$, choose q with $\varepsilon \leq \min\{1/2, \delta/(6m)\}$ and set

$$k_0 = \max\left\{ m(q+1),\ 2mq,\ 1,\ \left\lceil \frac{4mq\,\overline{H}}{\delta} \right\rceil \right\}. \tag{18}$$

Then $|\Xi_{m,k} - H| \leq \delta$ for *every* integer $k \geq k_0$. The lower error is at most $\delta/2$ and the upper error at most $2\delta/3$. This is a sufficient threshold, not the smallest one.

A computable upper bound on H does not require evaluating logarithms. If $c \geq 1$ and $Er \leq cr$, induction gives $M_m \leq c^m$, hence $H \leq m(c-1)$. The cutoff supplies another bound,

$$H \leq A_m(q) - 1 + (m-1)(R-1).$$

One may use the smaller of these two bounds. A cutoff used to bound H need not be the cutoff subsequently chosen for the accuracy tolerance.

4.2 An explicit all-powers tail

The following classical construction supplies (12) from positive normalized data. Set

$$\pi_i = \ell_i r_i, \quad \pi_* = \min_i \pi_i, \quad \eta = \min_{i,j} \frac{B_{ij}}{P_{ij}}, \quad \alpha = \eta/2, \quad \theta = 1 - \alpha.$$

The fixed-vector identities imply $0 < \eta \leq 1$, since each row of B_{ij}/P_{ij} has weighted average 1 with weights π_j. Let $T = (B - \alpha P)/\theta$. It is nonnegative, fixes the Perron vectors, and satisfies $TP = PT = P$. Induction gives

$$B^g = P + \theta^g (T^g - P).$$

From $T^g r = r$ and nonnegativity, $0 \leq (T^g)_{ij}/P_{ij} \leq 1/\pi_j$. Therefore a global relative tail is

$$\varepsilon_q = \theta^q/\pi_*, \qquad \left| \frac{(B^g)_{ij}}{P_{ij}} - 1 \right| \leq \varepsilon_q \quad (g \geq q). \tag{19}$$

When $B = P$, the exact bound $\varepsilon_q = 0$ is available for $q \geq 1$. For rational normalized matrices and a rational Perron frame, all minorization identities and the choice of q in (18) can be checked with exact rational arithmetic. General inputs require certified Perron data; a floating eigensolver is not such a certificate.

Corollary 6 (Quantitative convergence)**.** *For each fixed positive pair and fixed m,*

$$k \log(\rho_{m,k}/\lambda) = \log M_m + O(\log k/k).$$

Proof. Write $c_* = 1/\pi_*$ and $b = -\log\theta > 0$. Take $q = \max\{1, \lceil \log(c_* k)/b \rceil\}$, so $\varepsilon_q \leq 1/k$ and $q = O(\log k)$. For all sufficiently large k, both $k \geq m(q+1)$ and $k \geq 2mq$ hold. Equation (17) with fixed $\overline{H}$ gives the result. The constants depend on the fixed matrices and m. □

We do not claim that the logarithmic factor or these constants are optimal.

4.3 Rational stability tests

Put $A = A_m(q)$, $U = AR^{m-1}$ and $L = k - mq$. Exponentiating the finite-window bounds gives

$$\rho_{m,k}^{kL} \le \lambda^{kL} U^k (1+\varepsilon)^{mL}, \qquad \rho_{m,k}^{k} \ge \lambda^k \max\{1, (1-\varepsilon)A\}.$$

Thus a first right-hand side below 1 certifies exponential stability; a second right-hand side at least 1 excludes it. For rational data these comparisons are exact integer-power calculations. Otherwise the result may be inconclusive. These tests do not claim that the period-k witness is optimal or that the certified window is minimal.

5 Gap structure and its limitations

In dimension two, write the second eigenvalue of B as μ, where $|\mu| < 1$. Then $B^g = P + \mu^g(I - P)$, and, with $a = \ell^{\mathsf{T}} E r$ and $b_g = \ell^{\mathsf{T}} E B^g E r$,

$$b_g = a^2 + \beta \mu^g, \qquad \beta = \ell^{\mathsf{T}} E (I - P) E r.$$

The positive and negative cases for μ show that

$$M_2 = \max\{1, a^2, b_0, b_1\}. \tag{20}$$

For $\mu = 0$, the zeroth power is interpreted separately as $B^0 = I$. The finite list in (20) is not dimension independent.

Proposition 7 (Prescribed optimal pair gap)**.** *For every integer $t \ge 1$ there is a strictly positive pair in dimension $n = t + 1$ for which b_g is uniquely maximized, over all finite nonnegative gaps, at $g = t$, and its value exceeds the infinite-gap limit.*

Proof. Index coordinates by $0, \dots, n-1$. Let $R_{i,i+1 \bmod n} = 1$ be the cyclic permutation matrix and let $P = \mathbf{1}\mathbf{1}^{\mathsf{T}}/n$. Set

$$B = \tfrac{1}{2}(R + P), \quad v = \mathbf{1} + n e_0, \quad w = \mathbf{1} + n e_1, \quad E = v w^{\mathsf{T}}/(2n).$$

Take $r = \mathbf{1}$ and $\ell = \mathbf{1}/n$. All entries are positive and $a = 2$. Since $R^g e_0 = e_{-g \bmod n}$ and $B^g = P + 2^{-g}(R^g - P)$, direct multiplication gives

$$b_g = 4 + 2^{-g}\big(n \mathbf{1}_{\{g \equiv t \pmod n\}} - 1\big).$$

Outside the residue class t, this is less than 4. Inside that class it is greater than 4 and strictly decreases with g. The unique maximum is therefore $b_t = 4 + t/2^t > 4$. □

Choosing t outside any proposed fixed finite list defeats that list. This does not exclude dimension-dependent bounds or adaptive truncation, and it is not a statement about exact finite-k maximizing words.

6 A reset application at finite window sizes

The following elementary application of rank-one factorization [1] separates the asymptotic coefficient from finite-window optimality. It is not asserted as an independent novelty claim.

Proposition 8 (Rank-one success mode)**.** *Assume $B = P = r\ell^{\mathsf{T}}$ and set $e_j = \ell^{\mathsf{T}} E^j r$. If $e_1 \le 1$, $e_2 \ge 1$ and $e_3 \le e_2$, then for $k \ge 4$,*

$$\rho_{2,k} = \lambda e_2^{1/k}, \qquad \rho_{3,k} = \lambda e_2^{1/(k-1)}, \qquad M_2 = M_3 = e_2. \tag{21}$$

Proof. Every nonempty success run equals P. Thus $\ell^{\mathsf{T}} W r$ factors into the values e_j of the consecutive miss runs, including boundary runs. Under budget three these runs have length at most three. Delete single misses, and delete one endpoint of each triple. This preserves admissibility and does not decrease the scalar gain, since $e_1 \le 1$ and $e_3 \le e_2$. The remaining runs are pairs. Their starts are at least $k-1$ slots apart, or four misses would fit inside one k-window. At most $N/(k-1)+1$ pairs remain in N slots. Equation (5) proves the upper rate in the second formula. Repeating two misses followed by $k-3$ successes has period $k-1$, is admissible, and has normalized period spectral radius e_2. It attains the matching lower bound.

For budget two, delete single misses. Pair starts must be at least k slots apart: otherwise the last miss of one pair and both misses of the next fit in a k-window. The same bound and a pair repeated every k slots give the first formula. Finally $S_2 = \max(e_2, e_1^2) = e_2$ and $S_3 = \max(e_3, e_1 e_2, e_1^3) \le e_2$, proving the last equality. □

For example, take

$$B = \begin{pmatrix} 99/100 & 1/100 \\ 99/100 & 1/100 \end{pmatrix}, \qquad E = \begin{pmatrix} 1/10 & 1/10 \\ 20 & 1/10 \end{pmatrix}.$$

Here $r = \mathbf{1}$, $\ell^{\mathsf{T}} = (99/100, 1/100)$, and

$$e_1 = 399/1000, \qquad e_2 = 10349/5000, \qquad e_3 = 120797/100000.$$

Set $C = (9/10)B$ and $D = (9/10)E$. Under budget three, uniform exponential stability is equivalent to $\lambda^{k-1} e_2 < 1$. At $k = 7$ and $k = 8$ respectively, the exact comparisons are

$$\frac{5499882909}{5000000000} > 1, \qquad \frac{49498946181}{50000000000} < 1.$$

Thus eight is the first stable window among $k \ge 4$. The maximizing period is $k-1$. When $e_2 > 1$, the two budgets have different finite-k rates despite identical first-order coefficients.

7 Formal statement and verification scope

The accompanying Lean 4.32.2 development builds on the pinned mathlib library [11]. The declaration `MissedUpdates.Matrices.rare_miss_cluster_law` takes strictly positive real matrices on a nonempty finite index type, constructs their normalized Perron data, and proves the displayed limit for every fixed natural miss budget. The paper restricts to $m \ge 1$; the formal statement additionally covers $m = 0$. Definitions also allow small k, which does not change the eventual k-limit.

The formal norm is the elementwise maximum norm. Its two-sided comparison with the Perron gain is proved with constants independent of the word. The ordinary spectral radius is defined using the complex spectrum of the complexified real matrix; its equality to the positive normalizing eigenvalue is proved before converting it to a real number. Nonempty/bounded supremum sets and the necessary limsup boundedness are established explicitly. Gap order is reversed between

Mathematical object	Lean source modules	Semantic point checked
Window rule	`Protocol.lean`	Every starting slot of the infinite sequence; false means a miss
Chronological product	`Words.lean`, `Decode.lean`	Earliest action on the right; actual prefix recovered from canonical misses
Cluster budget	`Clusters.lean`, `Assembly.lean`	Derived lengths, separation and per-bin miss totals; no supplied growth bound
Perron normalization	`PerronConvergence.lean`, `PerronRadius.lean`	Data and convergence from positivity; full complex spectrum
Norm-growth rate	`NormGrowth.lean`, `Normalization.lean`	Literal supremum/limsup of norm roots, proved equivalent to gain growth
Coefficient	`GapSuprema.lean`	Genuine integer gaps and exact equality to $\max(1, S_1, \ldots, S_m)$
Finite cutoff	`Cutoff.lean`	Explicit finite domain, attained maximum and witness length
Quantitative bounds	`Quantitative.lean`, `QuantitativeError.lean`	Both finite-window inequalities and the all-later-windows accuracy implication

Table 1: Definition-to-proof map. Module names locate the code; full declaration names and types appear in the audit artifacts.

chronological words and left-to-right factors, and the formal product identity accounts for this reversal.

The combinatorial proof uses the sufficient occupied-bin count $\lfloor N/(k - mq) \rfloor + 1$ in place of the ceiling used above. The extra constant vanishes at the long-product rate. For the upper limit, the formal argument fixes a tail tolerance, lets k grow, and then shrinks the tolerance. The lower argument uses the period $k + a$ construction above. These are proof choices, not additional hypotheses on the model.

Perron eigenvector existence and finite-chain Dobrushin convergence are imported from the pinned HNBM source [5], not claimed as new results. The copied PF/MCMC files credit Matteo Cipollina in their headers; all 20 files remain byte-identical to the recorded upstream commit, with their original notices retained. This project supplies the normalization bridge, canonical word/matrix assembly, rate comparison and limit proof.

The quantitative structure `RelativeTail` requires both inequalities in (12) for every later power, together with $0 \leq \varepsilon < 1$. It does not assume the desired growth conclusion. Theorem 5, the cutoff comparison, (16), and an all-later-windows accuracy implication from the scalar conditions of (18) are kernel-checked. The automatic minorization construction (19) and Corollary 6 are given here with classical proofs but are not separately Lean-formalized. The exact rational tool validates its input frame, minorization identities and finite search; it must not be described as a verified floating-point front end for arbitrary matrices.

At this revision, the audit checks 201 exported theorem declarations using Lean trust level zero. Their dependency closures use only `propext`, `Classical.choice` and `Quot.sound`, with no custom mathematical axioms or proof placeholders. A separate consumer checks the exported theorem. Invalid controls test the protocol, scalar gain, limit coefficient, positivity assumption and quantitative window condition. These checks support the artifact; a human semantic review of the correspondence in Table 1 remains a separate pending gate.

8 Verification and publication status

The accompanying standard-library Python checker uses exact rational arithmetic for prescribed fixtures. It checks 408 admissible-word matrix/packing cases and 7,767 reset run-replacement cases, rejects three deliberately false assertions, and verifies finite instances of the prescribed-gap identities and coefficient enclosures. These finite checks support the calculations; they do not prove Theorem 1. The all-parameter argument is the classical proof above. The checker is fixture-specific, not a general validator for arbitrary matrix inputs.

The formal verification scope is specified in Section 7. The quantitative tool additionally checks exact rational tail identities, attained finite-cutoff values, conservative stability classifications and accuracy certificates. These are not tests of previously unknown matrix trajectories in place of proof.

The manuscript, code and proof scripts used AI assistance. Kernel checking does not establish novelty or replace author accountability. Authorship, affiliations, a fresh reader assessment and human specialist review remain unresolved. No reviewer has been appointed and no manuscript has been sent.

A targeted prior-art audit found no direct match in the inspected theorem statements, but this is not evidence sufficient to establish priority. Important original texts remained unavailable, including Zhao's 2019 work on constrained ergodic optimization for asymptotically additive potentials, Hertneck–Linsenmayer–Allgower's 2021 weakly-hard control paper, and the Gurvits–Samorodnitsky 2005 original attributed in the revision appendix of [1]. No result is attributed directly to an unread text. The exact 2019 Kozyakin–Kuznetsov–Chebotarev survey has now been read at its relevant statements and is cited directly. The earlier preprints [12, 4] were inspected separately, without identifying them with unread journal versions. The review materials retain the remaining access gaps and distinguish proposed reviewers from received assessments.

References

[1] Amir Ali Ahmadi and Pablo A. Parrilo. Joint spectral radius of rank one matrices and the maximum cycle mean problem. In *51st IEEE Conference on Decision and Control*, pages 731–733, 2012. Author-hosted published copy includes revision appendix.

[2] Maximilian Gaukler, Tim Rheinfels, Peter Ulbrich, and Günter Roppenecker. Convergence rate abstractions for weakly-hard real-time control. arXiv:1912.09871v2, 2020. Version dated 24 June 2020.

[3] Ihab Haidar, Yacine Chitour, Jamal Daafouz, Paolo Mason, and Mario Sigalotti. Stability criteria for singularly perturbed impulsive linear switched systems. arXiv:2507.02446v3, 2026. Version dated 6 February 2026.

[4] Michael Hertneck, Steffen Linsenmayer, and Frank Allgöwer. Stability analysis for nonlinear weakly hard real-time control systems. arXiv:2005.01345v2, 2020. Accepted IFAC version with proofs; not a substitute for the uninspected 2021 article.

[5] HNBM contributors. HNBM lean source repository: PF and MCMC modules. https://github.com/mkaxiomatic-ai/HNBM. Pinned commit 58b321e9c9b86448e4fdd897a428c8cef7297f5e; copied source headers credit Matteo Cipollina, 2025.

[6] V. S. Kozyakin, N. A. Kuznetsov, and P. Yu. Chebotarev. Consensus in asynchronous multiagent systems. III. constructive stability and stabilizability. *Automation and Remote Control*, 80(6):989–1015, 2019. Author-hosted published PDF; relevant block-frequency and spectral sections inspected.

[7] Victor Kozyakin. Matrix products with constraints on the sliding block relative frequencies of different factors. *Linear Algebra and its Applications*, 457:244–260, 2014. Inspected version: arXiv:1403.5050v4.

[8] Jun Liu and Mingqing Xiao. Rank-one characterization of joint spectral radius of finite matrix family. *Linear Algebra and its Applications*, 438(8):3258–3277, 2013. Inspected version: arXiv:1109.1356v8.

[9] Pierre Monmarché, Sebastian J. Schreiber, and Édouard Strickler. Impacts of tempo and mode of environmental fluctuations on population growth: Slow- and fast-limit approximations of Lyapunov exponents for periodic and random environments. *Bulletin of Mathematical Biology*, 87:81, 2025.

[10] Matthew Philippe, Ray Essick, Geir E. Dullerud, and Raphaël M. Jungers. Stability of discrete-time switching systems with constrained switching sequences. *Automatica*, 72:242–250, 2016. Inspected version: arXiv:1503.06984v3.

[11] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*. ACM, 2020.

[12] Yun Zhao. Conditional ergodic averages for asymptotically additive potentials. arXiv:1405.1648v1, 2014. Inspected as a distinct preprint; not identified with the uninspected 2019 journal version.